

GPRS Communication Protocol (Universal category)

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All data in this protocol is in the format of [Vendor*Device ID*Content Length*Content], of which the vendor ID always has two bytes, and the content is always four-byte ASCII code, with high order in front and low order behind, e.g.: 00A2 means the length is 162.

I. Terminal Sending Instructions

1. Link retention

The terminal sends:

[CS*YYYYYYYY*LEN*LK,number of steps, number of rolls, percentage of power]

Example: [3G*8800000015*000D*LK,50,100,100]

The platform replies:[CS*YYYYYYYYYY*LEN*LK]

Example: [3G*8800000015*0002*LK]

Description: The command is sent by interval, and a new connection will be established if the terminal fails to receive the data returned.

2. Reporting location data

The terminal sends:

[CS*YYYYYYYYYY*LEN*UD,location data (see Appendix I)]

Example:

[3G*2016001000*00BC*UD,120118,070625,A,22.570720,N,113.8620167,E,0.00,188.6,0.0,9,100,51,14188,0,00000010,6,255,460,0,9360,5081,156,9360,4081,129,9360,4151,128,9360,5082,127,9360,4723,122,9360,4082,120,0,22.4]

The platform replies: None

Description: The terminal will report the current location to the server.

3. Blind spot data transmission

The terminal sends:

[CS*YYYYYYYYYY*LEN*UD2,location data (see Appendix I)]

Example:

[3G*2016001000*00BD*UD2,120118,070625,A,22.570720,N,113.8620167,E,0.00,188.6,0.0,9,100,51,14188,0,00000010,6,255,460,0,9360,5081,156,9360,4081,129,9360,4151,128,9360,5082,127,9360,4723,122,9360,4082,120,0,22.4]

The platform replies: None

Note: Transmit the report data generated before logging in the platform.

4. Report alarm data

The terminal sends:

[CS*YYYYYYYYYY*LEN*AL, location data (see Appendix 1)]

Example:

[3G*2016001000*00BC*AL,120118,070625,A,22.570720,N,113.8620167,E,0.00,188.6,0.0,9,1
00,51,14188,0,00010010,6,255,460,0,9360,5081,156,9360,4081,129,9360,4151,128,9360,50
82,127,9360,4723,122,9360,4082,120,0,22.4]

The platform replies: [CS*YYYYYYYYYY*LEN*AL]

Example:

[3G*8800000015*0002*AL]

Description: The terminal will send the alarm information to the platform after an alarm is generated. If the terminal does not receive a reply, it will report again.

II. Platform Sending Instructions

1. Data upload interval setting

The platform sends:

[CS*YYYYYYYYYY*LEN*UPLOAD,time interval]

Example: [3G*8800000015*0009*UPLOAD,600]

The terminal replies:

[CS*YYYYYYYYYY*LEN*UPLOAD]

Example: [3G*8800000015*0006*UPLOAD]

Description: Set the interval for timing reporting of the terminal, and the unit is second.

2. Center number setting

The platform sends:

[CS*YYYYYYYYYY*LEN*CENTER,center number]

Example: [3G*8800000015*0012*CENTER,0000000000]

The terminal replies:

[CS*YYYYYYYYYY*LEN*CENTER]

Example: [3G*8800000015*0006*CENTER]

Note: Set the center number, and send SMS instructions through the mobile number.

3. Control password setting

The platform sends:

[CS*YYYYYYYYYY*LEN*PW,password]

Example: [3G*8800000015*0009*PW,111111]

The terminal replies:

[CS*YYYYYYYYYY*LEN*PW]

Example: [3G*8800000015*0002*PW]

Description: Set the password of the terminal, and add the password when a SMS

command is sent from a non-center number.

5. Monitoring

The platform sends:

[CS*YYYYYYYYYY*LEN*MONITOR]

Example: [3G*8800000015*0007*MONITOR]

The terminal replies:

[CS*YYYYYYYYYY*LEN*MONITOR]

Example: [3G*8800000015*0007*MONITOR]

Description: The terminal dials back the number of the center automatically.

6. SOS number setting

The platform sends:

[CS*YYYYYYYYYY*LEN*SOS,phone number, phone number, phone number]

Example: [3G*8800000015*0027*SOS,0000000000,0000000000,0000000000]

The terminal replies:

[CS*YYYYYYYYYY*LEN*SOS3]

Example: [3G*8800000015*0003*SOS]

Description: Set 3 SOS numbers in sequence.

7. Setting language and time zone

The platform sends:

[CS*YYYYYYYYYY*LEN*LZ,language, time zone]

Example: [3G*8800000015*0006*LZ,1,8]

The terminal replies:

[CS*YYYYYYYYYY*LEN*LZ]

Example: [3G*8800000015*0002*LZ]

Description: Set the terminal language and time zone.

8. SOS SMS alarm switch

The platform sends:

[CS*YYYYYYYYYY*LEN*SOSMS,0 or 1]

Example: [3G*8800000015*0008*SOSMS,0]

The terminal replies:

[CS*YYYYYYYYYY*LEN*SOSMS]

Example: [3G*8800000015*0006*SOSMS]

Description: Set whether to send SMS to SOS number after generating SOS alarm (0: off, 1: on).

9. Low battery SMS alarm switch

The platform sends:

[CS*YYYYYYYYYY*LEN*LOWBAT,0 or 1]

Example: [3G*8800000015*0008*LOWBAT,1]

The terminal replies:

[CS*YYYYYYYYYYY*LEN*LOWBAT]

Example: [3G*8800000015*0006*LOWBAT]

Description: Set whether to send SMS to the center number after a low battery alarm occurs (0: off, 1: on).

10. Version query

The platform sends:

[CS*YYYYYYYYYYY*LEN*VERNO]

Example: [3G*8800000015*0005*VERNO]

The terminal replies:

[CS*YYYYYYYYYYY*LEN*VERNO,version number]

Example: [3G*8800000015*0028*VERNO,G29_BASE_V1.00_2014.04.23_17.46.49]

Description: Query the version of terminal software.

11. Position now instruction

The platform sends:

[CS*YYYYYYYYYYY*LEN*CR]

Example: [3G*8800000015*0002*CR]

The terminal replies:

[CS*YYYYYYYYYYY*LEN*RESET]

Example: [3G*8800000015*0002*CR]

Description: Wake up the GPS module at the terminal immediately, and remain in

positioning status for a continuous period of time.

12. Power-off instruction

The platform sends:

[CS*YYYYYYYYYY*LEN*POWEROFF]

Example: [3G*5678901234*0008*POWEROFF]

The terminal replies:

[CS*YYYYYYYYYY*LEN*POWEROFF]

Example: [3G*5678901234*0008*POWEROFF]

Description: Power-off function.

13. Step counting time period setting

The platform sends:

[CS*YYYYYYYYYY*LEN*WALKTIME,time period, time period, time period]

Example: [3G*5678901234*002A*WALKTIME,8:10-9:30,10:10-11:30,12:10-13:30]

The terminal replies:

[CS*YYYYYYYYYY*LEN*ANY]

Example: [3G*5678901234*0008*WALKTIME]

Description: Set the time period range for step counting on.

14. Flip detection time period setting

The platform sends:

[CS*YYYYYYYYYYY*LEN*SLEEPTIME,time period]

Example: [3G*5678901234*0014*SLEEPTIME,21:10-7:30]

The terminal replies:

[CS*YYYYYYYYYYY*LEN*ANY]

Example: [3G*5678901234*0009*SLEEPTIME]

Description: Set the time period range of flip detection.

15. DND time period setting

The platform sends:

[CS*YYYYYYYYYYY*LEN*SILENCETIME,time period, time period, time period, time period]

Example:

[3G*5678901234*0037*SILENCETIME,21:10-7:30,21:10-7:30,21:10-7:30,21:10-7:30]

The terminal replies:

[CS*YYYYYYYYYYY*LEN*SILENCETIME]

Example: [3G*5678901234*000B*SILENCETIME]

Note: Set the DND time period range.

16. Find-watch instruction

The platform sends:

[CS*YYYYYYYYYYY*LEN*FIND]

Example: [3G*5678901234*0004*FIND]

The terminal replies:

[CS*YYYYYYYYYY*LEN*FIND]

Example: [3G*5678901234*0004*FIND]

Description: Ring for 1 minute while sending the command.

17. Alarm setting instructions

The platform sends:

[CS*YYYYYYYYYY*LEN*REMIND,alarm clock 1, alarm clock 2, alarm clock 3]

Example: [3G*5678901234*0018*REMIND,08:10-1-1,08:10-1-2, 08:10-1-3-0111110]

The terminal replies:

[CS*YYYYYYYYYY*LEN*REMIND]

Example: [3G*5678901234*0006*REMIND]

Description:Alarm time-switch-frequency -type.

18. Set a command for the phone book

The platform sends (1-5):

[CS*YYYYYYYYYY*LEN*PHB,number 1,name 1,number 2,name 2,number 3,name 3, number 4,name 4,number 5,name 5]

Example: [3G*5678901234*001B*PHB,313131,597D003100320033,,,,,,,,]

The terminal replies:

[CS*YYYYYYYYYY*LEN*PHB]

Example: [3G*5678901234*0003*PHB]

The platform sends (6-10):

[CS*YYYYYYYYYY*LEN*PHB2,number 6,name 6,number 7,name 7,number 8,name 8,
number 9,name 9,number 10,name 10]

Example: [3G*8800000015*001C*PHB2,313131,597D003100320033,,,,,,,,]

The terminal replies:

[CS*YYYYYYYYYY*LEN*PHB2]

Example: [3G*8800000015*0004*PHB2]

Description: The command is set for the phone book at the terminal.

19. Intercom function

The platform sends:

[CS*YYYYYYYYYY*LEN*TK,AMR format audio data]

The terminal replies:

[CS*YYYYYYYYYY*LEN*TK,receive result]

Audio data in ARM format needs to be unescaped:

0X7D 0X01 --> 0X7D

0X7D 0X02 --> 0X5B

0X7D 0X03 --> 0X5D

0X7D 0X04 --> 0X2C

0X7D 0X05 --> 0X2A

Receive result: 1—success

0- fail

The terminal sends:

[CS*YYYYYYYYYY*LEN*TK,AMR format audio data]

The platform replies:[CS*YYYYYYYYYY*LEN*TK,receive result]

Audio data in ARM format needs to be escaped:

0X7D --> 0X7D 0X01

0X5B --> 0X7D 0X02

0X5D --> 0X7D 0X03

0X2C --> 0X7D 0X04

0X2A --> 0X7D 0X05

Receive result: 1—success

0-fail

The terminal requests issuing an audio clip:

[CS*YYYYYYYYYY*LEN*TKQ]

III. Appendix:

Appendix 1: Position Data Description

Name	Example (ASCII code)	Description
Date	120414	(Day, month, year) 12, April, 2014
Time	101930	(Hour, minute, second) 10:19:30
Whether to locate	A	A: Yes, V: No

Latitude	22.564025	According to the definition of DD.DDDDDD format, this latitude value is: 22.564025
Latitude identification	N	N indicates north latitude, S indicates south latitude
Longitude	113.242329	According to the definition of DDD.DDDDDD format, this longitude value is: 113.242329
Longitude identification	E	E indicates east longitude, W indicates west longitude
Speed	5.21	5.21 km/h
Direction	152	The direction is at 152°
Altitude	100	Unit is meter
Number of satellites	9	Indicates the number of GPS satellites
GSM signal strength	100	Indicates the current GSM signal strength (0-100)
Power	90	Indicates the current power level percentage
Step count	1000	Step count is 1000
Number of rolls	50	Roll 50 times
Terminal status	00000000	Expressed as a hexadecimal string, the meaning is as follows: High 16bit means alarm, low 16bit means status Bit (start from 0) meaning (1 valid) 0 Low power 1 Out of the fence

		2 In the fence 3 Wristband status 16 SOS alarm 17 Low battery alarm 18 Out of fence alarm 19 In fence alarm 20 Low battery alarm
Number of base stations	4	Number of base stations reported, 0 means no base station information reported
Connect to base station ta	1	GSM delay
MCC country code	460	460 for China
MNC	02	02 for China Mobile
Area code of the connected base station	10133	Area code
Number of the connected base station	5173	Base station number
Signal strength of the connected base station	100	Signal strength

Location code of nearby base station 1	10133	Area code
Number of nearby base station 1	5173	Base station number
Signal strength of nearby base station 1	100	Signal strength
Location code of nearby base station 2	10133	Area code
Number of nearby base station 2	5173	Base station number
Signal strength of nearby base station 2	100	Signal strength
Location code of nearby base station 3	10133	Area code
Number of nearby base station 3	5173	Base station number
Signal strength of	100	Signal strength

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nearby base station 3		
...	...	...
Number of Wifi information	5	Number of Wifi (up to 5), sorted by signal strength
Wifi 1 name	rrr	Name of 1st wifi
Wifi 1 MAC address	1c:fa:68:13:a5:b 4	MAC address of 1st wifi
Wifi 1 signal strength	-61	Signal strength of 1st wifi
Wifi 1 name	abc	Name of 2nd wifi
Wifi 1 MAC address	1c:fa:68:13:a5:b 5	MAC address of 2nd wifi
Wifi 1 signal strength	-87	Signal strength of 2nd wifi
...	...	...
Positioning accuracy	6.5	Unit: m